Forecasting the Impact of the Federal Sentencing Guidelines

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The Sentencing Reform Act of 1984, which created the U.S. Sentencing Commission, required that the commission consider prison capacity in drafting sentencing guidelines. As part of the commission's efforts to meet this requirement, we developed a computer simulation capable of projecting the impact of the commission's guidelines on future sentences and prison populations. According to our projections, "straight" probation sentences will be reduced significantly under the guidelines. However, reducing the use of probation will generate only modest demands on the prisons, and then mostly for community corrections phased in over the course of 5 years. We also project that, while average time served for violent offenses will increase substantially, average time served for most property crimes will remain largely the same. Finally, while we forecast that federal prison populations will grow markedly by the end of this century, this is more a result of the Anti-Drug Abuse Act of 1986 and the career offender provision of the Comprehensive Crime Control Act of 1984 than a result of the guidelines.

The Sentencing Reform Act of 1984, which created the U.S. Sentencing Commission, authorized the commission to develop and promulgate federal sentencing guidelines. When delegating this authority, Congress realized that guidelines were a major innovation whose effects would extend beyond the sentencing process to touch investigator, prosecutor, and correctional practices. Congress was especially concerned with prison impact and directed the commission to consider the capacity of federal prisons when drafting guidelines.

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Although methods exist for forecasting state prison populations, those methods are based on aggregate population statistics and extensions of existing trends.[1] None satisfied the commission's need to project prison demands resulting from a major policy intervention that was bound to disrupt existing criminal case disposition patterns. Faced with this unique problem, we led an effort at the commission to develop a sophisticated computer simulation.[2]

The simulation led to some general conclusions: First, probation sentences that require no form of confinement will be reduced significantly. For especially serious crimes, such as drug offenses and crimes against persons, probationary sentences will decline dramatically. For other crimes, like property offenses, the proportion of sentences involving some form of probation will not change appreciably, although probation with a condition of confinement may be substituted for straight probation.

Second, average time served for violent offenses will increase substantially. For most property crimes, average time served will remain largely unchanged; for some property crimes (burglary and income tax fraud) the average time served will go up.

Third, federal prison populations will grow markedly by the end of this century, more as a result of the Anti-Drug Abuse Act of 1986 and the career offender provision of the Comprehensive Crime Control Act of 1984 than as a result of the guidelines. An important aspect of this growth will be the increased use of community corrections as an intermediate sentence between probation and prison.

The commission has reported its findings to Congress.[3] That report was necessarily brief, however, focusing on results rather than methods. In this article, we return to the prison impact issue, provide additional detail about methodology, and update and extend the results.

DATA

The impact methodology is simple in concept. We projected future prosecutions and convictions, estimated how long those defendants would have been incarcerated based on current sentencing and correctional practices, and compared the resulting estimated prison demands to those that likely would result from guidelines-induced changes. In application, the methodology was more complex, but it is

1. Blumstein, Sentencing Policies and their Impact on Prison Populations, in 2 *Research on Sentencing: The Search for Reform* (A. Blumstein, J. Cohen, S. Martin, & M. Tonry ed. 1983).

2. A number of staff members contributed to the development of the impact model. We would like to thank Peter Hoffman, David Lombardero, Michael Laskey, Shelley Matsuba, Phyllis Newton, Liz Phillips, Laura Schleich, Ronnie Scotkin, and especially Gerry Gaes (Bureau of Prisons) and Eric Simon. Additionally, Professor Arnold Barnett provided invaluable expert advice.

3. U.S. Sentencing Commission, *Supplemental Report on the Initial Sentencing Guidelines and Policy Statements* (1987).

clear from even this simplified rendition that we required detailed information about sentencing and correctional practices that predate the guidelines.

During 1983, the Administrative Office of the U.S. Courts, with the support of the Federal Probation Division, began compiling a comprehensive offender-based data file about sentencing practices. Known as FPSSIS (Federal Probation Sentencing and Supervision Information System), these data provided information about most felony and serious misdemeanor cases leading to convictions since the middle of 1983. FPSSIS is a rich source of information that includes a description of the offense, a characterization of the offender's background and criminal record, whether the offender was convicted by guilty plea or trial, and the sentence received.

FPSSIS formed the core of the commission's current practices data, but it was necessary to augment the file for two reasons. First, FPSSIS did not include some variables that were deemed relevant to guideline-based sentencing. Second, it allowed only very rough estimates of time served by convicted offenders. The impact model required precise estimates of time served for the narrow offense and offender categories stipulated by the guidelines, and FPSSIS was unable to satisfy that need.

To enrich the FPSSIS data, thereby solving the first problem, the commission drew a sample from the 40,000 defendants convicted during fiscal year 1985 (October 1, 1984–September 30, 1985). This sample consisted of 11,000 defendants and represented a diverse cross-section of the federal offender population; included were cases for street crime, white-collar offenses, drug offenses, and other federal crimes. The commission developed a detailed data collection form expanding on characteristics of the offense and offender, and had the Probation Division of the Administrative Office complete the form for the 11,000 cases.

The Probation Division provided 10,500 responses in addition to the corresponding presentence investigation reports, thus giving the commission quantitative information in the form of 10,500 computer records and qualitative information in the form of 10,500 presentence reports.

In response to the second problem, data from the Bureau of Prisons and the Parole Commission were added to FPSSIS, enabling the commission accurately to estimate time served by convicted offenders. For each of the 10,500 records, the Bureau of Prisons specified one of the following: the length of time the offender served in prison; the length of time the offender was scheduled to serve in prison if a parole date had been set; or the length of time the offender was expected to remain in prison according to rules that the bureau routinely employs to estimate release dates. When the bureau could not provide estimates, the commission relied on computerized Parole Commission records and, when these were unavailable, estimates of time served based on an analysis of prevailing release practices.

The FPSSIS data were too recent to provide adequate information about current probation and parole supervision practices and time served following a parole revocation. For this information, the commission relied on two sources: extant tabulations and commission-conducted statistical analyses based on supervision

histories and a sample of reports of revocation hearings conducted by the Parole Commission since 1977.

Collectively these data, fleshed out by a few other sources, enabled the research staff to complete extensive analyses on current sentencing practices. A comparison of these practices (as encapsuled by the analyses) with sentences prescribed by the guidelines foreshadowed the guidelines' projected impact on federal prisons. As we will show, while the guidelines implement several specific statutory requirements,[4] they conform roughly to current sentencing practices and are thus expected to have a modest impact on prisons.

This preliminary discussion notwithstanding, there are significant obstacles to projecting prison impact based on a simple comparison between current sentencing practices and the guidelines. Such a comparison provides no information about the number of offenders who will be sentenced for specific crimes, and requires an assumption that the extremely long sentences for certain defendants—drug dealers, career offenders, and violent criminals—will be applied in full, an occurrence that is unlikely in the face of significant incentives to plea bargain. The methodology applied in the commission's prison impact model overcomes these and other obstacles.

METHODOLOGY

To estimate prison impact, it is necessary to project future prison demands in the absence of guidelines, thereby establishing a "baseline." This already formidable task is complicated by two new laws, both of which are expected to increase the frequency and duration of prison terms. The Anti-Drug Abuse Act of 1986 raised the minimum and maximum prison terms for most convicted drug dealers; the career offender provision of the Crime Control Act of 1984 required lengthened prison time for certain repeat offenders. Both laws had to be incorporated into the guidelines, but it would be misleading to attribute the effect of their provisions to the guidelines themselves. Thus it was necessary to compare incarceration under the guidelines with a baseline projection based on the cumulative effect

4. A major objective was to classify prison impact into two categories: that attributable to commission policy decisions when Congress provided the commission with broad policy discretion, and that attributable to the commission's compliance when Congress narrowly confined the commission's authority. The classification is imprecise. For example, the dictates of the Anti-Drug Abuse Act are clear, and the commission had little choice in setting the sentences for drug dealers. It had somewhat more latitude in dealing with career offenders; still, it was required to increase terms for this group. Although our analysis distinguished the effects of the drug law and career offender provisions from the effects of the guidelines, it was impractical to differentiate the guidelines' impact from other congressionally imposed constraints on the commission. For example, the Crime Control Act abolishes parole, so a life sentence will result in a much longer prison stay than is typical currently. As another example, the Crime Control Act requires a "substantial term of incarceration" for defendants who "derive a substantial proportion of income [from] a pattern of criminal conduct." These and other similar aspects of the 1984 act constrained the development of guidelines; thus the "pure" prison impact of the guidelines is elusive.

of current sentencing and parole practices, current trends in prosecutions, the new drug law, and the career criminal provisions.

Current Sentencing Practices

As discussed earlier, we had detailed information about federal offenders who were sentenced during fiscal year 1985. To develop the baseline, we made two fundamental assumptions. First, within an offense category—like bank robbery—offenders and offenses observed during 1985 mirror offenders and offenses that would be observed after 1985. For example, if half the robbers used guns during 1985, half would use guns during 1990, half during 1995, and so on. Second, in the absence of guidelines, sentences imposed and time served by 1985 defendants are indicative of sentences and time that would be served by future offenders with similar offense and offender characteristics. As an example, if during 1985 embezzlers who stole $500,000 served 12 months in prison, we assumed that during 1990 (absent guidelines) embezzlers who steal a similar amount would serve the same length of time.

Conviction Trends

One obvious factor that influences any prison population forecast is the future level of crime. Except for short-term projections, however, no reliable method exists for predicting future federal crime rates. Even if future crime rates could be predicted with an acceptable degree of accuracy, historical data indicate a very weak relationship between changes in the number of offenses of a given type and changes in the contemporaneous level of federal prosecutions. Such discrepancies result partially from the exercise of discretion by U.S. attorneys, who may refer cases to state or local authorities, or simply decline prosecution for policy reasons.[5]

Discrepancies between changes in specific crime rates and prosecutions may also arise because of altered enforcement priorities. To the extent that these priorities are set by the administration in office, predicting future priorities involves predicting the outcomes of future elections and the policies pursued by successive administrations.

Rather than speculate about the future political climate, we adopted two different assumptions regarding future prosecutions. One presumes "low-growth" in prosecutions and the other "high growth." In the low-growth projections, we assumed that the average year-by-year increase from 1982 through 1986 would continue only through 1989. This assumption reflects the fact that the present administration, whose policies determined the growth pattern for 1982 through 1986, will have

5. U.S. Department of Justice, *U.S. Attorneys' Written Guidelines for the Declination of Federal Criminal Laws* (1979).

no direct influence on prosecutions after 1989. We assumed that criminal caseload would grow at a constant yearly growth rate of 1% thereafter.

In the high-growth projections, we assumed that the annual rate of growth in the number of criminal prosecutions for the next 5 years (1987 through 1991) would equal the average annual growth rate for the preceding 5 years (1982 through 1986), a period of exceptionally high growth. The growth rate for the remainder of the period was chosen so that total growth over the period from 1987 to 2002 would equal total growth during the previous 15-year period, 1971 to 1986, also a period of relatively high growth.

The two growth paths were applied to the 1985 cohort, effectively *projecting this cohort foward in time*. If, for example, there were 1000 forgers in 1986, and the growth path indicated that forgery cases would grow by 10% per year, we projected 1100 convicted forgers for 1987, 1210 for 1988, and so on. Basic to understanding the impact methodology is grasping how the 1985 cohort was used to simulate future caseloads.[6]

Plea Negotiations

Plea negotiation practices are another crucial element in prison population forecasts, because more than 85% of all federal defendants enter guilty pleas. We made two assumptions: (1) that, as we projected the 1985 cohort forward in time, the same defendants who entered guilty pleas in the past would enter guilty pleas in the future; and (2) that the percentage reduction received currently for a guilty plea (30 to 40% reduction in time served) would prevail in the future.

Parole Revocations

Not all future prison demands will result directly from sentences imposed by the court. Some defendants who were sentenced to probation during 1985 will

6. Any growth assumption necessarily has an element of arbitrariness about it, and it is possible to postulate and justify a scenario in which the growth rate is even lower than 1%. For example, during the period 1976 to 1980, the number of convictions for drug offenses, forgery, weapon offenses, rape, and robbery all fell by roughly 50%. Thus recent historical precedent might have been consistent with even lower projections.

Given congressional sentiment as expressed in the Anti-Drug Abuse Act, however, it appears unreasonable to assume that drug case prosecution patterns in the near future could replicate the trends of the 1976–1980 period. This is particularly significant because drug offenders constitute the single largest block of federal prisoners. In addition, the number of fraud cases increased from fewer than 1000 in 1970 to more than 5000 in 1984, producing the second largest block of federal prisoners. Since 1970, the number of fraud cases has declined only in 1979 and 1985. Robbers comprise the third largest block of federal prisoners. Again, the long-term growth is significant, from fewer than 300 robbery cases in 1961 to approximately 2300 in 1976. After the decline that occurred during the late 1970s, the number of robbery cases has stabilized at approximately 1200 per year. Based on long-term trends, our low-growth scenario is modest indeed. It implies that the number of cases would grow by approximately 16% over a 15-year period when, in fact, cases that result in the majority of prison sentences have grown by about 50% between 1971 and 1986. Even if the 1% growth rate assumption in the low-growth scenario is an overestimate, however, it would be several years before there would be any significant difference in prison populations.

have their supervision status revoked and, as a result, will be sentenced to prison. Also, some 1985 defendants who will be paroled will have their parole status revoked, resulting in additional prison time. We had to account for these secondary consequences of sentencing.

Our approach was to assign each 1985 cohort member who was expected to be released on parole both a probability of having his parole revoked and an additional prison term following a revocation. These assignments were based on a statistical analysis of actual supervision histories, which we assumed reflected patterns that would persist into the future (absent guidelines). The results were incorporated into the simulation.

Due to data limitations, it was not possible to conduct a statistical analysis of defendants who had their probation revoked. Because this group creates relatively few demands for the prisons, we ignored its effect, excluding probation revocations from the simulation.

Drug Law

The Anti-Drug Abuse Act of 1986 greatly increased most of the punishments for violating the nation's drug laws and extended the laws to cover some special situations. Because implementation of the drug law coincided with development of the impact model, we were unable to rely on actual observations to determine how the drug law would influence the baseline. We were forced instead to simulate sentencing under the new drug law.

The Anti-Drug Abuse Act stipulates minimum and maximum terms based on the amount of drugs involved in the offender's crime. Because drug amount was included in the 1985 cohort data, we used it to determine the minimum and maximum terms "at risk," that is, the shortest and longest sentences that could be served based on the drug amount. The basic approach was to compare time actually served by members of the 1985 cohort to the minimum and maximum terms required by the new drug law. When the defendant served less than the new drug law minimum, the computer model assumed that he would have served the minimum had the new drug law been applicable at the time he was sentenced. (The drug law increased maximum terms, so no defendants sentenced prior to the drug law served more time that the drug law maximums.) Otherwise, the model assumed that under the drug law he would serve exactly the sentence that he served according to the FPSSIS data.

This basic approach was modified in two regards. First, we assumed that the minimum terms would not constrain plea bargaining; specifically, we assumed that a defendant could negotiate for a sentence reduction that would be equivalent (in terms of percentage reduction in time served) to the sentence reduction that was typical for the 1985 cohort. Second, the Anti-Drug Abuse Act puts no constraints on sentences when the defendant cooperates with authorities, so in such cases (as identified in the 1985 data) we assumed that time served would be equivalent to what it was during 1985.

Career Offender Provision

Inclusion of the career offender provision was straightforward. The Crime Control Act of 1984 required the commission to set prison terms for certain repeat offenders (those convicted of violent crimes and selling drugs who had two or more prior convictions for similar crimes) "at or near the maximum term authorized [by law]." Thus the basic approach was to determine the minimum sentence required and the maximum sentence allowed by the guidelines, and then assume that the defendant would be sentenced to the minimum if his actual time was less than the minimum, to the maximum if his actual time was greater than the maximum, and to his actual time otherwise. These results were adjusted for plea-bargain concessions, using the same assumptions that were employed for drug cases.

Sentences Under the Guidelines

The next step was the "heart" of the impact analysis. To assess the impact of the guidelines on the federal prison population, the simulation "resentences" each member of the 1985 cohort using the guidelines' provisions. Each offender's data was examined to determine his conviction offense, his criminal history, and other circumstances of his crime to assess the sentence he was "at risk" of receiving under the guidelines. The guidelines prescribe a sentence range, so the at-risk maximum was the top of the range and the at-risk minimum was the bottom of the range.

The at-risk sentence was reduced by a specific percentage (depending on the offense type) when the defendant entered a guilty plea. These percentage reductions were determined from analyses of current plea negotiation practices. The at-risk sentence was reduced to the amount of time actually served by the 1985 cohort member when there was substantial cooperation with the government. We assumed that probation would be the guidelines sentence when the guidelines allowed probation as a choice (after the plea-bargain concession) and the 1985 cohort member had actually received probation. Otherwise, we assumed that the defendant would serve the midpoint of the guideline range, minus 15% when allowed by good-time provisions.

Delay in Application of the Guidelines

Not everyone convicted after November 1, 1987, will be sentenced under the guidelines. To be subject to the guidelines, the defendant must have been convicted of a crime *that was committed after November 1*. When we projected the 1985 cohort forward, we also projected forward the date that the defendants committed their crimes, and we simulated an application of the guidelines only when the projected crime commission date was later than November 1, 1987.

Sensitivity Analysis

Accuracy when measuring the effect of major changes was of utmost importance to the commission, but the projections were carried out in the presence of many imponderables. Because guideline-induced charges are likely to be substantial, and because future reaction by federal judges and others is unknown, we modified the simulation to accommodate several alternative scenarios.

We modified the basic scenario, discussed above, in three ways. First, as discussed earlier, we adopted alternative low- and high-growth paths, and compared projections for each. Second, we had to predict judicial reactions to the guidelines. Basically, we assumed that judges would depart infrequently from the guidelines. However, the reasonableness of this assumption is arguable, so we adopted alternatives that simulated various degrees of departure. Third, we had to model plea negotiations. The basic approach, as discussed already, was to assume that defendants will enter guilty pleas under the guidelines at the same rate as now, and that the percentage sentence reduction now exchanged for guilty pleas will approximate the reductions to occur in the future. Different assumptions, some of which allow greater sentence concessions and some of which allow lesser concessions, were introduced into the model. In total, prison population forecasts were derived for almost 20 different possible scenarios. The results under these various scenarios provide upper and lower bounds on the estimate of prison impact.

RESULTS

The simulation was directed toward answering questions about the guidelines' influence on sentencing practices and the resulting demands on federal prisons. Some of our findings are summarized next.

Guidelines Reduce Straight Probation Sentences

One of the effects of the guidelines will be to reduce straight probation. In Table 1 the percentages of offenders that currently receive straight probation and the corresponding percentages that are projected to receive straight probation under the guidelines are depicted in columns 1 and 2.[7] "Straight" probation is probation without any conditions of confinement. The last three columns show the percentage of defendants under current practices and the guidelines that receive probation with a condition of intermittent confinement, community confinement, or a split sentence.

Under current practice, probation terms that require some confinement include probation with jail as a condition of probation, split sentences, and mixed sentences.

7. These projections include the impact of the Anti-Drug Abuse Act of 1986 and the career offender provision of the Comprehensive Crime Control Act of 1984.

TABLE I Probation Terms and Split Sentences by Offense Type

	Percentage of Defendants Receiving Probation and Split Sentences				
	Straight Probation		Probation with Confinement and Split Sentences		
Offense Type	Under Current Practice	Under Guidelines	Under Current Practice	Under Guidelines (Probation with Custody)	Under Guidelines (Split Sentence Equivalent)
Person offenses	31.3%	14.7%	10.5%	3.1%	7.7%
Robbery	18.0%	4.0%	8.0%	1.0%	1.0%
Burglary	64.0%	43.0%	10.0%	3.0%	10.0%
Property offenses	60.1%	33.1%	15.2%	17.5%	18.1%
Drugs	20.8%	5.1%	13.0%	1.3%	6.9%
Fraud	59.0%	24.0%	18.0%	20.0%	21.0%
Income taxes	57.0%	13.0%	25.0%	25.0%	30.0%
Firearms	37.0%	9.0%	15.0%	17.0%	16.0%
Immigration	41.0%	30.0%	27.0%	8.0%	28.0%
Total	42.4%	18.9%	15.4%	10.4%	14.3%

Notes:

1. The term *straight probation* means probation without any period of confinement.
2. Under current practice, probation terms with confinement consist of probation terms with any form of incarceration: probation with jail as a condition, split sentences, and mixed sentences.
3. The term *probation with custody* means probation with a condition requiring a period of intermittent confinement or residence in a community treatment center.
4. The term *split sentence equivalent* means a sentence under the guideline §5C2.1(c)(3) or (d)(2).
5. The category labeled *person offenses* includes homicide, assault, rape, and kidnapping.
6. The category labeled *property offenses* includes embezzlement, forgery, larceny, property destruction, counterfeiting, and auto theft.
7. The category labeled *burglary* also includes trespass.

The guidelines create two new types of sentences similar in form to a split sentence. For minimum sentences of no more than 6 months, the defendant may receive a term of probation that includes a period of intermittent or community confinement. For minimum sentences of no more than 10 months, the defendant's sentence may be "split" between incarceration and community confinement.

As Table 1 indicates, under the guidelines straight probation will be reduced. Currently 42.4% of all convicted defendants receive straight probation. After the guidelines are implemented, only 18.9% are projected to receive this sentence. For some crimes, the impact is even more dramatic. For example, while in income taxes cases, 57% of persons convicted of tax evasion presently receive probation, this is projected to fall to 13% under the guideline.

The guidelines were designed to reduce the disparity in sentencing as well as increase the effectiveness of the sentencing process. Hence the guidelines decrease the proportion of defendants who receive sentences that do not include any confinement and eliminate disparately long prison terms. We project that the certainty of confinement will be dramatically increased under the guidelines.

For a number of especially serious crimes, probationary sentences (in any form) are projected to decline greatly under the guidelines. For crimes against the person, currently 41.8% of those convicted receive some form of probation; this is projected to decline to 25.5% after the guidelines are implemented. Likewise, the incidence of probation and split sentences for robbery will decline from 26.0 to 6.0% and for drug offense from 33.8 to 13.3%. Under the guidelines, probation in any form will be rare for any of these especially serious crimes.

For some other crimes, the overall proportion of sentences involving probation will not change appreciably under the guidelines, but confinement as a condition of probation will be required more frequently. Specifically, for property crimes, the percentage of defendants who receive straight probation will drop from 60.1 to 33.1%. By contrast, the percentage of offenders who receive probation terms requiring some confinement will increase to 35.6 from 15.2%. Overall, an estimated 75.3% of property offenders currently receive some form of probation, including split sentences, and an estimated 68.7% will continue to do so under the guidelines.

Guidelines Will Increase Average Time Served Somewhat

Average time served will increase from 15.3 months given current practices to 28.7 months under the guidelines. However, most of this increase will be concentrated in a few crimes and for most crimes time served will remain essentially unchanged. Estimates of the average time served currently and projections of future sentencing practices are set forth in Table 2.

When estimating the impact of the guidelines on the average length of sentences, we attempted to isolate the effects of the Anti-Drug Abuse Act of 1986 and the career offender provision of the Comprehensive Crime Control Act of 1984. The 1986 drug law significantly increases the minimum and maximum prison terms for most drug offenses. Additionally, the Comprehensive Crime Control Act of 1984 required the commission to adopt guidelines that would ensure that certain

TABLE II Time Served Under Current Practice and Projected Time Served Under the Drug Law, Career Offender Provision, and Guidelines

	Average Number of Months Served (Imprisonment)			
	Present Practice		Future Practice	
Offense Type	Current Practice	Drug Law	Career Offender	Guidelines
Robbery	44.8	—	74.1	74.6
Person	37.7	—	53.3	76.7
Drugs	23.1	48.0	56.7	57.6
Firearms	14.1	—	—	15.2
Burglary	7.7	—	8.9	10.8
Fraud	7.0	—	—	8.2
Property	6.8	—	—	6.6
Immigration	5.7	—	—	5.2
Income tax	5.5	—	—	8.6

Notes:
1. Average time served is based on sentences for all offenders. Offenders not sentenced to imprisonment are treated as having zero months' imprisonment.
2. The average time served reported in column 2 (drug law) and column 3 (career offender provision) is for all offenders, not only offenders subject to the new drug law and the career offender provision.
3. Imprisonment includes confinement in prison, jail, or a community corrections facility.

repeat offenders—those 18 or older who are convicted of a violent crime or drug offense and who had been convicted of two or more such crimes previously—receive sentences "at or near the maximum term authorized." In accordance with this provision, the commission established for these offenders guideline sentences close to the statutory maximums.

In Table 2, Column 1 indicates the average time served by those convicted of various offenses. These averages include probationers who contribute zero time unless incarcerated as a condition of probation. The second column projects average time served after implementation of the drug law. The third column reports projected average time served under both the drug law and the career offender provision. Finally, the last column projects average time served after implementation of (1) the new drug law, (2) the career offender provision, and (3) the guidelines.

As an illustration, consider the entries for "Drugs" in Table 2. Prior to the new drug law, defendants convicted of drug violations served an average of 23.1 months. This average is projected to increase to 48.0 months after the drug law is fully implemented. It is projected to increase to 56.7 months when the drug law and the career offender provision are both taken into account. And factoring in the guidelines will increase this to 57.6 months. As this example demonstrates, the impact on average sentence lengths produced by the new drug law and the career offender provision will, in many cases, be much larger than the impact of the guidelines themselves. In fact, only for burglary are the guidelines themselves projected to have a larger effect on average sentence length than the career offender provision.

In most cases when neither the drug law nor the career offender provision is applicable, average sentence lengths are not altered radically. For these crimes (firearms, fraud, property, and immigration) the projected reduction in straight probation will be the major impact of the guidelines. More defendants will be confined, but many will be confined for a shorter term than is presently the case, and their confinement will frequently be in community-based or other minimum security facilities.

Income tax is the one crime not subject to either the drug law or the career offender provision where the guidelines themselves are projected to have a major impact on average sentence length. Here, the average sentence is projected to increase from a current level of less than 6 months to approximately 9 months. For income tax cases, not only will probation be reduced and hence the probability of confinement be increased, but the overall amount of confinement will also be increased. This increase in sentence severity results from treating income tax evasion in a manner similar to other frauds resulting in comparable dollar losses.

The Guidelines Alone Will Not Have a Major Impact on Prison Population

Although straight probation is reduced by the guidelines and time served on average is somewhat longer, the precise effect of these changes in sentencing practices on overall prison demand is not obvious. For example, a significant increase in the sentence for a crime rarely prosecuted will have little effect on prison demand. By contrast, a small increase or decrease in the average sentence for a frequently prosecuted crime may have a substantial impact.

The commission has projected prison demand for the years 1992, 1997, and 2002. The projections for 1992 are believed to be the most accurate because prosecution policies are unlikely to change significantly over this time period. This is especially true over the next 3 years, during which the present administration will continue to influence prosecution policies. The 10-year projections merit somewhat less confidence. Nevertheless, they provide a general impression about prison demand over a longer time period. The 15-year projections are necessarily very speculative and are presented only so the long-term relative effects of the Anti-Drug Abuse Act, career offender provision, and guidelines can be considered.

To estimate the impact of the guidelines, the Anti-Drug Abuse Act, and the career offender provision on the federal prison system, it was necessary first to establish as a "baseline" a projection of future prison demand in the absence of any change in current sentencing practices. We calculated future prison populations under both a low-and a high-growth scenario. Table 3 provides 15-year prison population projections under the low-growth assumption. Projections are given for both existing and future sentencing practices.

Even before considering the effect of these new statutes, prison population under this low-growth scenario is projected to increase from 41,000 to 54,000 in 1992, to 58,000 in 1997, and to 62,000 in 2002. In other words, we might

TABLE III Projected Growth in Prison Population: Low-Growth Scenario

	Projected Prison Population by Year			
	1987	1992	1997	2002
Baseline	41,000	54,000	58,000	62,000
With drug law	N/A	64,000 (+19%)	82,000 (+41%)	90,000 (+46%)
With career offender provision	N/A	66,000 (+2%)	87,000 (+6%)	100,000 (+10%)
With guidelines	N/A	69,000 (+5%)	89,000 (+2%)	102,000 (+2%)

Notes:

1. The baseline projections for 1987–1989 are based on the average annual growth for the years 1982–1986. Thereafter, criminal cases were assumed to grow at a 1% rate.
2. The percentage change is the increase relative to the previous estimate. The percentage change for the drug law is the change relative to the baseline. The percentage change for the career offender provision is the change relative to the baseline plus the drug law. The percentage change for the guidelines is the change relative to the baseline plus the drug law and the career offender provision.

expect more than a 50% increase in prison population in the absence of any change in sentencing practices.

As Table 3 indicates, the new drug law will cause a substantial increase in future prison populations. We project that it will increase the prison population to 64,000 in 1992, to 82,000 in 1997, and to 90,000 in 2002. Given that the drug law is already in effect and a part of the sentencing structure, the projected increase from 41,000 in 1987 to 82,000 in 1997 makes a significant point: A doubling of inmates over the next decade might be expected *even under a low-growth assumption and even if the commission did not exist.*

Now if we take into account the career offender provision in addition to the drug law, prison population is projected to increase to 66,000 in 1992, to 87,000 in 1997, and to 100,000 in 2002. Adding the effects of the guidelines into the analyses raises the projected prison population to 69,000 in 1992, to 89,000 in 1997, and to 102,000 in 2002. This last calculation emphasizes another important point: *The incremental effect of the guidelines is relatively modest, only about 5% in 1992, 2% in 1997, and 2% in 2002.*[8]

Aside from the impact that is created by the growth in prosecutions, the most significant factor in creating a demand for prison capacity will be implementation of the new drug law. While the career offender provision is projected to have a small effect on population in the near term, its absolute and relative impact will increase over time. On the other hand, at least over the next decade, the guidelines are likely to have relatively minor impacts on prison population.

The 15-year "high-growth" projections are set forth in Table 4. Under this scenario, baseline prison population is expected to increase dramatically from

8. When drug law violators are excluded altogether from the calculations, the marginal impact of the guidelines is somewhat higher. Excluding drug cases, the guidelines will result in an estimated 10% increase in prison demands for 1992 and a 6% increase in prison demands for 2002. When we also exclude career offenders, the impact is between 6 and 9%.

TABLE IV Projected Growth in Prison Population: High-Growth Scenario

	Projected Prison Population by Year			
	1987	1992	1997	2002
Baseline	41,000	60,000	75,000	92,000
With drug law	N/A	71,000 (+18%)	105,000 (+39%)	133,000 (+45%)
With career offender provision	N/A	72,000 (+2%)	110,000 (+5%)	144,000 (+8%)
With guidelines	N/A	76,000 (+6%)	113,000 (5%)	149,000 (+4%)

Notes:

1. The baseline projections for 1987–1991 are based on the average annual growth for 1982–1986. Projections for 1992–2002 are based on the average annual growth that reproduces the same 15-year percentage growth for 1987–2002 that was observed for 1971–1986.
2. The percentage change is the increase relative to the previous estimate. The percentage change for the drug law is the change relative to the baseline. The percentage change for the career offender provision is the change relative to the baseline plus the drug law. The percentage change for the guidelines is the change relative to the baseline plus the drug law and the career offender provision.

41,000 currently to 60,000 in 1992, 75,000 in 1997, and 92,000 in 2002. If prosecutions actually grow at this high-growth rate over the 15-year period, we could expect more than a doubling of prison population without any change in sentencing practices.

These high-growth projections suggest that, given current trends in prosecutions, future demands on the federal prisons will exceed present prison capacity by significant proportions. While we treat these projections as the upper limit of future prison demand, they are not beyond the range of experience. Over the last 15 years, the population in federal prisons has grown from more than 20,000 to more than 40,000.

Under this high-growth scenario, the drug law is projected to have a substantial effect on future prison population. Because of the drug law, prison population is projected to increase by more than 18% in 1992, 39% in 1997, and 45% in 2002. The effects of the guidelines are modest in comparison. Specifically, the additional effect attributable to the guidelines is 6% in 1992, 5% in 1997, and 4% in 2002.[9] Although the impact of the guidelines is somewhat larger under this scenario than under the low-growth scenario, it is still modest.

Reducing the Incidence of Probation Will Generate Modest Demands on the Prisons

As we stated earlier, the significant reduction in the incidence of straight probation terms may be the most important innovation attributable directly to

9. After excluding all drug law violators from the calculations, the marginal impact attributable to the guidelines is 11 to 12%. After excluding drug law violators and career offenders from the calculations, the marginal impact of the guidelines is 9 to 12%.

commission policy. In the abstract, this policy may seem likely to have an explosive effect on the prisons. In fact, this is not true.

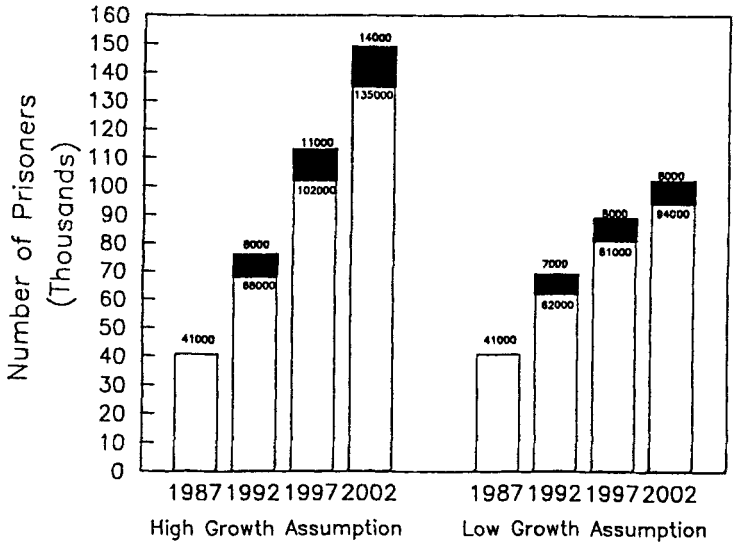
Figure 1 illustrates the impact on the prisons of reducing straight probation terms. The total height of the bars indicates the total demands on the prisons. The shaded portions of the bars represent prison time served by defendants who would receive straight probation given current sentencing practices, but who are projected to receive some confinement under the guidelines.

The result of reducing straight probation, when compared to the results of other processes that generate prison demands, is moderate. Nevertheless, the result is greater than the total impact of the guidelines, as represented in Tables 3 and 4. The explanation for this difference makes an important point: The guidelines reduce sentence disparity by decreasing both the number of sentences that are too lenient (inappropriate probation terms) and the number that are too severe (unjustifiably long prison terms) without greatly changing average sentences.

Timing of the Expected Impact

The guidelines became effective on November 1, 1987, but not everyone convicted after November 1 will be sentenced under the guidelines—only defendants whose crimes were committed after November 1. Therefore, implementation of the guidelines will be gradual.

The speed of the guidelines' phase-in was relevant for judging how much of a "shock" the guidelines would impose on the courts and corrections during the



Note: Shaded portions of the bars represent prison time served by defendants who would receive probation given current sentencing practices.

Figure 1. Projected prison populations highlighting growth attributable to reducing probation.

first few months, when the guidelines were novel. Using the method described earlier, we assumed a high-growth forecast and projected the number of guidelines cases on a month-by-month basis for the first 5 years following implementation.

The resulting projections of the number of defendants who will be sentenced under the guidelines are reported in Table 5. The table shows the projected number of defendants by offense category who will be sentenced under the guidelines during specified months after the guidelines take effect. Two additional rows report the total number of defendants per specified month who will be sentenced under the guidelines, as well as the total percentage of all defendants sentenced during a specified month after the guidelines take effect who will be sentenced under the guidelines.

It was especially difficult to predict the numbers of guideline cases during the first few months of the guidelines' operation. We suspected that the "date of commission" might become a negotiable item during plea bargaining, that some delay would occur because the guidelines were new, that judges might postpone hearing guidelines cases, and that some judges and courts would find the guidelines unconstitutional (as some have done). These facts all operate to reduce the number of guidelines case projections based on the methodology used here.

Nevertheless, these estimates were valuable approximations, showing as they did that the guidelines could not have a serious, instantaneous disruptive effect because—at least for the early months after implementation—few guidelines; cases would be processed.[10] These findings were used by the commission (and others, such as the probation division) to plan for the future (regarding issues such as training and monitoring) and to argue before Congress that implementation of the guidelines should not be delayed beyond November 1, 1987.[11]

Use of Community Corrections Will Increase

It is difficult to make accurate projections about community corrections usage, given the uncertainty that exists regarding how judges will choose to apply the guidelines. This uncertainty is greatest for "split sentences," because the court has sole discretion to divide the confinement period between prison and a community-based alternative. The uncertainty is still significant for probation terms, when community confinement is a condition of probation, because this is the guideline sentence that probably makes the greatest change in existing sentencing practices. We are uncertain as to how judges will respond.

10. The projections overestimated the number of guideline cases. We had projected between 5000 and 7000 defendants by June 1987. In fact, there appear to have been about 2200 defendants sentenced, and 3400 pending, during this 8-month period.

11. Concern had been expressed in several quarters, including the federal judiciary, that implementation should be delayed because the courts were not prepared to deal with guidelines on November 1, 1987. The commission's position, which prevailed in public debate, was that the gradual phase-in of the guidelines provided ample time to prepare the court for the guideline regime.

TABLE V Projected Number of Defendants Who Will Be Sentenced Under the Guidelines

	Months After Implementation of the Guidelines								
	1	2	3	4	5	6	7	8	9
Homicide	0	0	1	3	4	5	6	6	7
Sex offenses	0	1	3	4	5	6	7	8	9
Larceny	6	25	41	68	86	100	116	131	141
Embezzlement	2	8	17	26	36	44	54	61	67
Auto theft	1	2	3	5	7	8	10	12	13
Robbery	2	8	21	36	47	55	61	66	70
Forgery/ counterfeiting	9	15	38	63	78	84	93	103	108
Bribery	0	0	1	2	3	3	4	5	5
Drugs	23	50	159	326	458	546	609	650	712
Fraud	13	26	45	73	112	134	149	170	188
Firearms	5	8	18	31	45	56	65	71	76
Immigration	30	101	147	174	189	192	199	202	204
Tax fraud	2	3	5	7	8	11	11	14	15
Other	32	69	116	192	332	428	479	571	623
Total	125	316	616	1011	1409	1672	1864	2071	2237
Percentage of defendants	3%	8%	16%	25%	36%	42%	47%	52%	56%

Note: The columns report the number of defendants who are expected to be sentenced under the guidelines during specified months after the guidelines take effect. Dates are rounded to the nearest month. The augmented FPSSIS case data were used to develop these estimates. For every defendant in the data file, we examined the date that the defendant ended his participation in the offense and compared that date to the date he was sentenced. We assumed (1) that the elapsed time between when the defendant's participation ended and when he was sentenced, as recorded in the augmented FPSSIS data, will be the same for defendants who will be sentenced after the guidelines go into effect, and (2) that the number of future criminal cases will follow a high-growth path, as is defined in the technical report.

Despite this uncertainty, it is clear that community corrections facilities will be used more heavily under the guidelines. This projected increase can be attributed to the anticipated decline in the use of straight probation and an anticipated increase in the length of sentences that might be served in community corrections facilities.

Based on the simulation, we concluded that had the offenders in the 1985 cohort been sentenced under the guidelines, somewhat more than 2500 would have received community or intermittent confinement. Approximately another 4000 likely would have received split sentences. We estimated that these 6500 offenders could have required 1600 to 3600 man-years of community corrections facility space depending on the length of sentences and the proportions of split sentences allocated to community facilities. This contrasts with the approximately 800 man-years currently used for direct commitments from court and for commitments as a condition of probation.

These figures are long-run estimates, which apply only after all defendants sentenced in federal courts are sentenced under the guidelines. Some judges and

Months After Implementation of the Guidelines								
10	11	12	13–18	19–24	25–36	37–48	49–60	over 60
8	8	8	54	58	121	124	125	125
11	12	12	88	90	182	182	182	182
149	163	166	1275	1452	3171	3356	3474	3567
76	81	88	671	739	1676	1759	1801	1816
14	15	16	132	153	342	358	366	369
72	75	77	495	515	1053	1072	1083	1091
118	122	129	916	1013	2298	2477	2552	2582
6	6	7	52	60	150	168	180	195
750	806	830	5673	6020	13079	13668	14075	14318
210	218	233	1793	2154	5318	6008	6336	6566
80	88	99	755	857	1853	1911	1937	1964
204	205	205	1266	1287	2586	2591	2602	2610
16	16	17	145	158	539	728	1016	1253
630	642	684	4788	4990	10510	10720	10808	10961
2342	2458	2570	18103	19545	42878	45123	46537	47599
59%	62%	65%	76%	82%	90%	95%	98%	100%

others have expressed concern that the guidelines would put a large and almost immediate demand on limited community corrections bed space. Thus we also projected the demand for community corrections bed space for the period just after the guidelines were implemented (excluding bed spaces required for offenders in transition between prison and release). By 1989, we project that a total of 1144 beds will be required for community or intermittent confinement and the portion of split sentences spent in community confinement facilities. This figure is expected to grow to about 2000 by 1992, an estimate that is within the 1600 to 3600 beds for the long-run estimates.

We also estimated the number of defendants sentenced to community confinement.[12] By 1992, it is projected that almost 6000 defendants per year will be sentenced to community corrections, close to the 6500 defendants expected to be sentenced to community corrections when the guidelines are fully operational.

Several conclusions can be drawn from this analysis. First, the immediate demand on community corrections is likely to be modest, because the guidelines will not be in full force until after 1992. Second, better projections (based on actual experience with sentences imposed under the guidelines) can be made well before the demands on the Bureau of Prisons have peaked.

12. Sentences that involve community corrections are typically less than a year, so the total bed space required for community corrections is less than the total number of offenders assigned to these beds.

OTHER LIMITATIONS; SENSITIVITY ANALYSIS

The simulation is based on an assumption that judges rarely depart from the guidelines. This assumption is questionable.[13] Judges are allowed to depart from guideline sentences for factors not adequately considered in the guidelines, provided they explain in writing their reasons for departure. While the commission does not expect departure from the guidelines to occur with great frequency, it is not known how often they will occur, which direction they will take, and how large they will be. Thus the discretion to depart from the guidelines, which is integral to the legislation that established the commission, creates uncertainty about the ultimate impact of the guidelines on actual sentences.

Future plea negotiations are another problem that clouds prison projections.[14] Plea negotiations are an indispensable part of federal criminal justice administration. Recognizing the functional role of plea bargaining, the guidelines attempt to structure the bargaining process, not eliminate its practice. The environment in which plea bargaining will occur must change under the guidelines, but the nature and extent of the changes are unknowable, creating additional uncertainty for projecting future sentencing practices.

Using sensitivity analysis, the commission's impact projections were adjusted to account for the uncertainty inherent in projecting future prison demands in the face of major policy innovations. All of the results reported in this article are somewhat affected by alternative assumptions adopted for the sensitivity tests, but as we report in the Supplementary Report, the major conclusions are substantively unaffected.

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13. Given the experience of states with determinate sentencing laws and sentencing guidelines, as well as that of the federal Parole Commission, the assumption that departures will occur is a reasonable one. See Minnesota Sentencing Guidelines Comm'n, The Impact of the Minnesota Sentencing Guidelines: Three Year Evaluation (1984); U.S. Parole Comm'n, Report of the United States Parole Commission: October 1, 1980 to September 30, 1983 (1984); Kramer & Lubitz, Pennsylvania Sentencing Reform: The Impact of Commission Established Guidelines, 31 Crime & Delinq. 481 (1985); D. Carrow & J. Feins, Guidelines Without Force: An Evaluation of the Multijurisdictional Sentencing Guidelines Field Test (1985).

14. Some studies at the state level indicate that compulsory sentencing guidelines and determinate sentencing laws may change the nature of guilty pleas or the rate at which they are entered. See generally J. Casper, The Implementation of the California Determinate Sentencing Law (1981); Minnesota Sentencing Guidelines Comm'n, The Impact of the Minnesota Sentencing Guidelines: Three Year Evaluation (1984); J. Cohen and M. Tonry, Sentencing Reforms and Their Impacts in 2 *Research on Sentencing: The Search for Reform* (A. Blumstein, J. Cohen, S. Martin, & M. Tonry ed. 1983); S. Clarke, North Carolina's Determinate Sentencing Legislation: An Evaluation of the First Year's Experience (1983); R. Morelli, The Impact of Five-Year Mandatory Sentencing on Three Pennsylvania Trials Court: A Preliminary Analysis (1985) (Paper presented at the 1985 Annual Meeting of the American Society of Criminology); M. Tonry, Sentencing Reform Impacts, National Institute of Justice, Washington, D.C. (1987); R. Hendberg and N. Holten, The Impact of Florida's Sentencing Guidelines on the Processing and Disposition of Criminal Cases (1985); D. Carrow and J. Feins, Guidelines Without Force: An Evaluation of the Multijurisdictional Sentencing Guidelines Field Test (1985).

Against the backdrop of these various problems, one might nevertheless fear that any projections of impact will be far too speculative to be trusted. But decision makers do not have the luxury of adopting this view. Failure to project the impact of the guidelines on sentences and on prison capacity as realistically as possible could convert an imprecise decision-making process into a wholly arbitrary one.